

YEAR 10 SCIENCE (EXTENSION)
PHYSICAL SCIENCES
REVISION - TEST 2

1. A dragster at Kwinana Motorplex uniformly accelerates from rest and covers the 400 m track with a final velocity of 309 kmh⁻¹.

(a) Calculate the acceleration of the vehicle.

$$v = 85.8 \text{ ms}^{-1}$$

$$u = 0 \text{ ms}^{-1}$$

$$a = ?$$

$$t = ?$$

$$s = 400 \text{ m}$$

Take forwards as +ve.

$$v^2 = u^2 + 2as$$

$$\Rightarrow a = \frac{v^2 - u^2}{2s}$$

$$= \frac{(85.8)^2 - 0}{2(400)}$$

$$= \underline{9.20 \text{ ms}^{-2} \text{ forwards}}$$

(b) How long does it take the dragster to cover the 400 m?

$$v = u + at$$

$$\Rightarrow t = \frac{v - u}{a}$$

$$= \frac{85.8 - 0}{9.20}$$

$$= \underline{9.33 \text{ s}}$$

2. The highest "ridiculous" basketball shot is currently set from the top of Maletsuyane Falls in Lesotho. Assume the initial vertical velocity is zero when it is released and ignore any sideways movement.

(a) If it takes 6.41 s to fall to the ring, what was the ball's vertical velocity when it reached the ring?

$$v = ?$$

$$u = 0 \text{ ms}^{-1}$$

$$a = 9.80 \text{ ms}^{-2}$$

$$t = 6.41 \text{ s}$$

$$s = ?$$

↓ +ve

$$v = u + at$$

$$= 0 + (9.80)(6.41)$$

$$= \underline{62.8 \text{ ms}^{-1} \text{ down}}$$

- (b) Calculate the height of the falls.

$$\begin{aligned}s &= vt + \frac{1}{2}at^2 \\ &= 0 + \frac{1}{2}(9.80)(6.41)^2 \\ &= \underline{201\text{ m}}\end{aligned}$$

No direction needed as displacement was not asked.

3. An ES-3A Shadow jet moving at 240 kmh^{-1} relative to an aircraft carrier lands on its flight deck and the tail hook snags the arresting wire, which stops the jet in 2.30 s .

- (a) Determine the deceleration experienced by the pilot.

$$v = 0\text{ ms}^{-1}$$

$$u = 66.7\text{ ms}^{-1}$$

$$a = ?$$

$$t = 2.30\text{ s}$$

$$s = ?$$

Take forwards as +ve.

$$a = \frac{v-u}{t}$$

$$= \frac{0-66.7}{2.30}$$

$$= \underline{-29.0\text{ ms}^{-2}\text{ backwards}}$$

- (b) How far does the jet move while stopping?

$$s = ut + \frac{1}{2}at^2$$

$$= (66.7)(2.30) + \frac{1}{2}(-29.0)(2.30)^2$$

$$= \underline{76.7\text{ m forwards}}$$

4. In the Tour de France, a cyclist descending down the Col de Portet was moving at 110 kmh^{-1} as he approached a sharp bend. He braked evenly and decelerated at 3.38 ms^{-2} to make the turn at 40.0 kmh^{-1} . How long did the rider brake? *Take forwards as +ve.*

$$\begin{aligned} v &= 11.1 \text{ ms}^{-1} \\ u &= 30.6 \text{ ms}^{-1} \\ a &= -3.38 \text{ ms}^{-2} \\ t &= ? \\ s &= ? \end{aligned}$$

$$\begin{aligned} a &= \frac{v-u}{t} \\ \Rightarrow t &= \frac{v-u}{a} \\ &= \frac{11.1 - 30.6}{-3.38} \\ &= \underline{5.77 \text{ s}} \end{aligned}$$

5. The Shinkansen high-speed train in Japan decelerated at 2.0 ms^{-2} for 35 s as it approached Kobe station and came to a stop.

- (a) What was the velocity of the train before it started braking?

$$\begin{aligned} v &= 0 \text{ ms}^{-1} \\ u &= ? \\ a &= -2.0 \text{ ms}^{-2} \\ t &= 35.0 \text{ s} \\ s &= ? \end{aligned}$$

$$\begin{aligned} &\text{Take forwards as +ve.} \\ v &= u + at \\ \Rightarrow u &= v - at \\ &= 0 - (-2.0)(35.0) \\ &= \underline{70.0 \text{ ms}^{-1} \text{ forwards}} \end{aligned}$$

- (b) How far did the train travel while decelerating?

$$\begin{aligned} s &= ut + \frac{1}{2}at^2 \\ &= (70.0)(35.0) + \frac{1}{2}(-2.0)(35.0)^2 \\ &= \underline{1225 \text{ m forwards}} \end{aligned}$$

6. Complete these statements about Newton's three laws of motion.

(a) An object remains at rest, or will not change its speed or direction, unless ...

acted on by an external, unbalanced force.

(b) When an unbalanced force acts on an object, the mass of an object affects ...

the size of the acceleration or deceleration experienced.

(c) For every action, there is ...

an equal and opposite reaction.

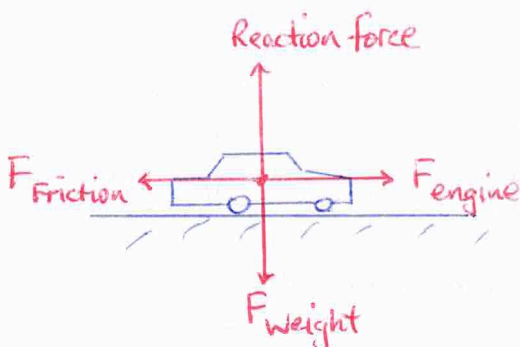
7. Complete the following table.

Force	Mass	Acceleration
100 N	<i>20 kg</i>	5 m/s ²
<i>1.2 N</i>	0.300 kg	4 m/s ²
1.2 N	2 kg	<i>0.6 m/s²</i>

8. Explain why the following statement is **false**.

A car travelling along a straight road has no forces acting on it.

Draw a diagram showing all of the forces acting on the car as part of your answer.



• There are 4 forces acting at all times on the car.

9. Explain why it is dangerous to have loose objects inside a moving car when it brakes sharply or is involved in an accident.

- When the car brakes sharply or stops suddenly, the loose objects have inertia.
- This will carry them forwards with the original speed of the car, which could injure the passengers if they are hit.

10. Use Newton's Second Law and the equations of motion to calculate answers for the following.

- (a) A 1400 kg car accelerates at 3.0 ms^{-2} . What size force is needed to cause this acceleration?

$$F = ?$$

$$m = 1400 \text{ kg}$$

$$a = 3.0 \text{ ms}^{-2}$$

Take forwards as +ve.

$$F = ma$$

$$= (1400)(3.0)$$

$$= \underline{4200 \text{ N forwards}}$$

- (b) A force of 160 N causes an object to accelerate at 2.0 ms^{-2} . What is the object's mass?

$$F = 160 \text{ N}$$

$$m = ?$$

$$a = 2.0 \text{ ms}^{-2}$$

Take forwards as +ve.

$$F = ma$$

$$\Rightarrow m = \frac{F}{a}$$

$$= \frac{160}{2.0}$$

$$= \underline{80.0 \text{ kg}}$$

(c) A force of 210 N acts on a mass of 70 kg moving at 5.0 m/s for 8.0 s in a straight line.

(i) What is the final velocity of the mass?

$$F = 210 \text{ N}$$

$$m = 70.0 \text{ kg}$$

$$a = ?$$

Take forwards as +ve

$$F = ma$$

$$\Rightarrow a = \frac{F}{m}$$

$$= \frac{210}{70.0}$$

$$= 3.0 \text{ ms}^{-2} \text{ forwards}$$

$$v = ?$$

$$u = 5.0 \text{ ms}^{-1}$$

$$a = 3.0 \text{ ms}^{-2}$$

$$t = 8.0 \text{ s}$$

$$s = ?$$

$$v = u + at$$

$$= 5.0 + (3.0)(8.0)$$

$$= \underline{29.0 \text{ ms}^{-1} \text{ forwards}}$$

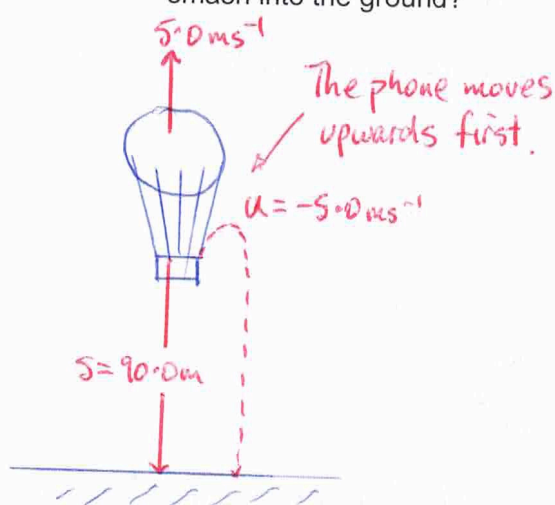
(ii) How far does the mass move while the force is acting?

$$s = ut + \frac{1}{2}at^2$$

$$= (5.0)(8.0) + \frac{1}{2}(3.0)(8.0)^2$$

$$= \underline{136 \text{ m forwards}}$$

11. A hot-air balloon is rising steadily at 5.0 ms^{-1} over a paddock near Northam on a clear, cold morning. A passenger, keen to get a good photo, leans over the side of the basket with her phone. Unfortunately, she is bumped by a fellow passenger and drops her phone. At that moment, the balloon was 90.0 m above the ground. How long did it take for the phone to smash into the ground?



↓ we $V =$

$$u = -5.0 \text{ ms}^{-1}$$

$$a = 9.80 \text{ ms}^{-2}$$

$$t = ?$$

$$s = 90.0 \text{ m}$$

consider the whole motion.

$$v^2 = u^2 + 2as$$

$$= (-5.0)^2 + 2(9.80)(90.0)$$

$$\Rightarrow v = 42.3 \text{ ms}^{-1}$$

$$v = u + at$$

$$\Rightarrow t = \frac{v - u}{a}$$

$$= \frac{42.3 - (-5.0)}{9.80}$$

$$= \underline{4.83 \text{ s}}$$

12. A car is travelling down the highway at 90 kmh^{-1} when a kangaroo bounces onto the road ahead. The driver takes 1.2 seconds to react and take his foot off the accelerator before braking hard.

- (a) How far will the car travel during the time the driver is reacting? This is called the reaction distance. (Hint - this is constant speed.)

$$v = 25.0 \text{ ms}^{-1}$$

$$s = ?$$

$$t = 1.2 \text{ s}$$

Take forwards as +ve.

$$v = \frac{s}{t}$$

$$\Rightarrow s = vt$$

$$= (25.0)(1.2)$$

$$= \underline{30.0 \text{ m forwards}}$$

- (b) The car's brakes can slow the car at a rate of 6.0 ms^{-2} . How long will it take for the car to come to a complete stop?

$$v = 0 \text{ ms}^{-1}$$

$$u = 25.0 \text{ ms}^{-1}$$

$$a = -6.0 \text{ ms}^{-2}$$

$$t = ?$$

$$s = ?$$

$$v = u + at$$

$$\Rightarrow t = \frac{v - u}{a}$$

$$= \frac{0 - 25.0}{-6.0}$$

$$= \underline{4.17 \text{ s}}$$

- (c) What force must be exerted by the brakes to stop the car if it has a mass of 1400 kg?

$$F = ?$$

$$m = 1400 \text{ kg}$$

$$a = -6.0 \text{ ms}^{-2}$$

$$F = ma$$

$$= (1400)(-6.0)$$

$$= \underline{-8400 \text{ N backwards}}$$

- (d) Calculate how far the car travels while braking to a stop.

$$s = ut + \frac{1}{2}at^2$$

$$= (25.0)(4.17) + \frac{1}{2}(-6.0)(4.17)^2$$

$$= \underline{52.1 \text{ m forwards}}$$

- (e) Determine the total stopping distance of the car from the moment the driver sees the kangaroo.

$$s_{\text{total}} = s_{\text{reaction}} + s_{\text{braking}}$$

$$= 30.0 + 52.1$$

$$= \underline{82.1 \text{ m forwards}}$$

13. (a) Explain the difference between **mass** and **weight**. Give an example to help your explanation.

MASS: amount of matter in an object.

WEIGHT: force due to gravity on an object.

- (b) The acceleration due to gravity on Jupiter is 22.5 ms^{-2} . Calculate the weight of a 450 kg object on Jupiter.

$$F_w = ?$$

$$m = 450 \text{ kg}$$

$$g = 22.5 \text{ ms}^{-2}$$

$$F_w = mg$$

$$= (450)(22.5)$$

$$= \underline{10,130 \text{ N}}$$

- (c) If, on Earth, you can jump vertically to a height of 50.0 cm, how high could you jump on Jupiter (if it was solid)? Justify your answer by some form of calculation. (Hint: Consider what your initial velocity is on Earth.)

EARTH

$$v = 0 \text{ ms}^{-1}$$

$$u = ?$$

$$a = 9.80 \text{ ms}^{-2}$$

$$t = ?$$

$$s = -0.5 \text{ m}$$

$$\downarrow \text{ +ve } v^2 = u^2 + 2as$$

$$\Rightarrow u^2 = v^2 - 2as$$

$$= 0 - 2(9.80)(-0.5)$$

$$\Rightarrow u = 3.13 \text{ ms}^{-1} \text{ up.}$$

JUPITER

$$v = 0 \text{ ms}^{-1}$$

$$u = -3.13 \text{ ms}^{-1}$$

$$a = 22.5 \text{ ms}^{-2}$$

$$t = ?$$

$$s = ?$$

$$v^2 = u^2 + 2as$$

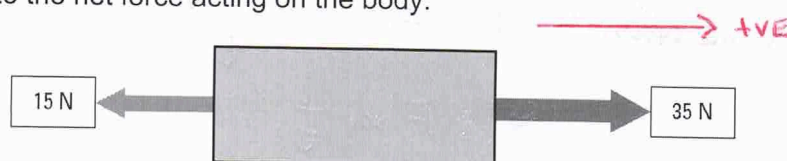
$$\Rightarrow s = \frac{v^2 - u^2}{2a}$$

$$= \frac{0 - (3.13)^2}{2(22.5)}$$

$$= \underline{-0.218 \text{ m up}}$$

14. Consider the forces acting on a body of mass 40 kg resting on a frictionless surface.

- (a) Calculate the net force acting on the body.



$$\Sigma F = 35 - 15$$

$$= \underline{20 \text{ N right}}$$

- (b) Calculate the acceleration of the body.

$$\Sigma F = 20 \text{ N}$$

$$m = 40 \text{ kg}$$

$$a = ?$$

$$\Sigma F = ma$$

$$\Rightarrow a = \frac{\Sigma F}{m}$$

$$= \frac{20}{40}$$

$$= \underline{0.50 \text{ ms}^{-2} \text{ right}}$$

15. What is the difference between a **primary safety feature** and a **secondary safety feature**?

PRIMARY: helps prevent an accident happening.

SECONDARY: helps prevent or reduce injury to passengers during an accident.

16. By increasing the time over which a collision occurs, the force of the impact on a car and its passengers can be decreased.

List and describe at least **two safety features** of a car that decrease the force of impact in this way.

AIR BAG: collapses in a controlled way to decelerate a passenger over a longer time.

CRUMPLE ZONES: collapse in a controlled way to spread energy away from the passengers and decelerate the car slowly.

Should mention: $F = ma = \frac{m\Delta v}{t}$

$$\Rightarrow F \propto \frac{1}{t}$$

- Increase $t \Rightarrow F$ is smaller.

17. You studied a car safety feature as part of this topic.

(a) Name the safety feature and describe how it acts to protect passengers.

- Name the feature.
- Describe how it works.

(b) State which of Newton's Law's applies to this safety feature and explain the Physics principles involved.

- State the appropriate law(s).
- Explain how inertia (1st law), force felt (2nd law) and/or action/reaction (3rd law) is applied.